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EXAMINER
VOLKOV, ALEXANDER ALEXANDROVIC

ART UNIT	PAPER NUMBER
1677	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/280,426	Applicant(s) Allebrandt et al.	
	Examiner ALEXANDER VOLKOV	Art Unit 1677	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 14 May 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-12 and 16-18 is/are pending in the application.

5a) Of the above claim(s) 8-12 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 and 16-18 is/are rejected.

8) ☑ Claim(s) 1,5 and 16-17 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

Claims 1-12 and 16-18 were pending. Claims 8-12 were withdrawn. Claims 1 and 7 are amended.

Claims 1-7 and 16-18 are examined herein.

Claim Objections

Claims 1, 5, 16, and 17 are objected to because of the following informalities:

Claims 1, 5, 16, and 17 contain abbreviations MET, GDF15, CCL22, TNFRSFl1, ANGPT2, and ReLA NF-KB. Abbreviations should be completely spelled out in their first occurrence. The abbreviations were initially spelled out in claim 1(a) but were deleted in the amended claim 1.

Appropriate correction is required.

Withdrawn Rejections

The objection to the specification is withdrawn in view of specification amendments.

The rejections of claim 7 under 35 U.S.C. 112(b) is withdrawn in view of claim amendments.

The rejections of claims 1-7 and 16-18 under 35 U.S.C. 103 are withdrawn in view of claim 1 amendments. Claim 1 amendments necessitated a new prior art search and new grounds of rejection have been found in view of newly found prior art reference. The amended claim 1 is rejected under 35 U.S.C. § 103 as unpatentable over **Gerstein** in view of **Caldwell**.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL. —The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor

of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 1-7 and 16-18 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. **This is a written description rejection.**

Claim 1 is directed to determining, diagnosing, and/or prognosing an individual's risk of suffering from cardiovascular disease the method comprising: a. detecting, in a sample obtained from an individual, the expression level, amount and/or activity of two or more biomarkers; and b. comparing the expression level, amount, and/or activity of the biomarker with a reference from a healthy control population.

Claims 5 and 16-17 are similar to claim 1, but recite at least 3, 5, and 10 of the biomarkers respectively.

Claims 1, 5, and 16-17 are interpreted broadly as requiring at least 2, 3, 5, and 10 of the biomarkers respectively in any combination.

Nature of the invention – the invention is directed to diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease using measured biomarker levels.

State of the prior art & level of predictability in the art. The art of using biomarkers for diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease is highly unpredictable. Some prior art references teach biomarker trends contradicting those of instant invention. For example, **Dragut et al**. (Atherosclerosis, 235e239) teach that increased plasma levels of TNF alpha is associated with increased cardiovascular risk (e239, col. 1, last par.), which contradicts claim 2 reciting that a decrease in TNF alpha levels is indicative of an individual having a higher risk of suffering from cardiovascular disease or a negative prognosis.

Amount of direction and examples provided by the inventor.

The specification is silent with respect to specific data related to recited biomarkers and evidence of correlations with cardiovascular outcomes. For example, the specification fails to provide evidence of corresponding p-values, which is the key parameter characterizing a biomarker for its suitability in diagnostic applications. No biomarkers and corresponding p-values are disclosed. No combinations of the biomarkers and corresponding p-values for these combinations are disclosed.

Most of the disclosure is about network-related subject matter and computational analysis. For example, some figure titles:

Figure 1 shows after the disease network identification;

Figures 2A-2C and Figures 8A-8F show the networks of molecular entities that were prioritized in the overconnectivity analyses;

Figures 3A-3E show that the molecular signatures of population sub-types relate to CVD progression;

Figures 4A-4C show cox-regression results presented as Hazard rations (HR) with confidence intervals;

Figures 9A-9C show Kaplan-Meier survival estimates by patient clusters for the measured CV outcomes in Caucasian and Latinos.

The specification provides only two examples.

Example 1 - identification of biomarkers and biomarker networks. However, the bulk of this example is a disclosure of genotype-related subject matter and SNPs related to non-elected claims 8-12. For example: section "Genotyping quality control and population stratification analysis" (pg. 95-96) and section "Genome-wide association analyses" (pg. 96-97).

Example 2 - detection of SNPs associated with CVD risk – this example is entirely related to the non-elected claims.

The specification fails to demonstrate actual expression levels, amounts, or activities of the recited biomarkers for the healthy control population and the individuals suffering from cardiovascular diseases. The disclosure makes references to data not present in the specification. For example: Supplementary Table S1 (Patient Stratification, pg. 102), Supplementary Table S2 (ORIGIN GWAs results, pg. 105), Supplementary Table S3 (CVD disease network

prioritization, pg. 105), Supplementary Table S5 (last line, pg. 105). Page 112 lists titles of all six Supplementary Tables, but the tables are not present in the specification.

The prior art reference of **Gerstein et al**. (IDS; Circulation. 2015 Dec 15; 132(24) :2297-304) is the most closely related reference to instant disclosure. However, **Gerstein** fails to teach diagnostic potentials/performances of each of the biomarkers recited in claim 1, both as individual biomarkers and as combinations of the biomarkers.

Additionally, **Gerstein** teaches a set of 10 biomarkers for the cardiovascular composite outcome "Ten novel biomarkers were identified that were each independent determinants of the 1405 CV composite outcomes of nonfatal myocardial infarction, nonfatal stroke, or CV death" (section "Clinical perspective", last page). However, **Gerstein** fails to teach these 10 biomarkers as independent biomarkers, instead they are presented as a singular cardiovascular composite outcome or a set. Fig. 1B supports the 10 biomarkers as a set "Similar curves based on the fifths of the risk probability estimated from a model that that includes the baseline clinical variables plus the levels of the 10 identified biomarkers in Table 3" (Fig. 1B legend) – Fig. 1B does not show any individual biomarker.

Finally, **Gerstein** teaches only a fraction of the claimed biomarkers: N-terminal pro-B-type natriuretic peptide, trefoil factor 3, apolipoprotein B, angiopoietin-2, a-2-macroglobulin, glutathione S transferase a, growth/differentiation factor 15, and hepatocyte growth factor receptor. **Dragut et al**. (Atherosclerosis, 2014. 235e239) and **Lorbeer et al**. (Heart. 2015 Feb;101(3):178-84) teach angiopoietin-2 and RORA. The remaining 10 biomarkers of claim 1 are not disclosed as cardiovascular biomarkers.

Finally, claims 1, 5, and 16-17 recite measuring activity of the biomarkers. The specification fails to disclose measured activity of any of the biomarkers.

Claims 2-7 and 16-18 are rejected because they depend from rejected claim 1.

Based on the above findings, one of ordinary skill in the art would conclude that Applicant did not have possession of the claimed invention.

The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION. —The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-7 and 16-18 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 1 recites in step (b) comparing the expression level, amount and/or activity of the biomarkers MET, GDF15, CCL22, TNFRSF11, ANGPT2, and ReLA NF-KB but the same biomarkers are not recited in step (a) where the expression level, amount and/or activity of the biomarkers is detected. It is unclear how the expression level, amount and/or activity of the biomarkers can be compared to healthy controls if their expression level, amount and/or activity are not detected.

For the purpose of compact prosecution and applying prior art under 35 U.S.C. 102 and 103, the instant claim is interpreted as reciting the biomarkers MET, GDF15, CCL22, TNFRSF11, ANGPT2, and ReLA NF-KB previously recited in step (a) and deleted in the current amendment.

Claims 2-7 and 16-18 are rejected because they depend from rejected claim 1.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims 1-7 and 16-18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a naturally occurring correlation, without significantly more.

Claim 1 is directed to a process, which belong to the four statutory categories.

The claim is related to a method of determining, diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease, which is categorized as a naturally occurring correlation, and therefore it is a judicial exception.

Step (b) of the claim recites comparing the measured expression level, amount, and/or activity of the biomarker with a reference from a healthy control population. The limitation falls into the “mental process” groupings of abstract ideas and therefore is a judicial exception.

Claim 1 recites additional elements of detecting, in a sample obtained from an individual, the expression level, amount and/or activity of two or more biomarkers. The additional element of "detecting" is an insignificant extra-solution activity that amounts to mere data gathering necessary to apply the judicial exception.

When considering the elements in combination, the claim as a whole does not integrate the recited exceptions into a practical application. Claim 1 recites administering therapeutic agents: statin, beta blocker, blood thinning agent, nitrates, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists, calcium channel blockers and diuretics are not intended to treat any cardiovascular disease events (step (c)). However, for the administering step to be considered as a practical application of diagnostic method the recited therapeutic agents must be particular, which is not the case with the amended claim 1.

Specifically, claim 1 recites diagnosing a "cardiovascular disease (CVD) event" without subdividing known CVD events into groups with common treatment options. The specification fails to provide a clarification. As such, the method of claim 1 is intended to diagnose all known CVD events as a risk factor of any CVD event and treat the individual with a list of conflicting medications.

For example, administration of blood thinning agents to a patient with stroke caused by a burst of a blood vessel is going to harm a patient. The specification fails to disclose biomarkers capable of differentiating different cardiovascular disease events. Bundling all cardiovascular disease events into one CVD group fails to make administration of the recited therapeutic agents a practical application of the disclosed diagnostic method due to conflicting medications required by different cardiovascular disease events.

Therefore, claim 1 does not recite any practical applications of the recited method and is directed to the judicial exception.

Additional elements of measuring the biomarker levels and considered insignificant extra-solution activities are recited at a high level of generality and have been recognized as routine laboratory techniques. The techniques for measuring biomarker levels are known in the art. For example, it was routine and conventional to measure the same biomarker levels in the ORIGIN trial using the LUMINEX 100/200 platforms (**Gerstein et al**. (IDS; Circulation. 2015 Dec 15; 132(24) :2297-304); pg. 2298, col. 2, par. 2). Therefore, the claim as a whole does not

amount to significantly more than the recited exception, because there are no additional elements, or combination of additional elements, that add an inventive concept to the claim.

The dependent claims 2-7 and 16-18 fail to add additional elements, or combination of additional elements, that contribute to an inventive concept to the claim. For example: claims 2-4 recite directions of the biomarker levels correlating with a higher risk of suffering from cardiovascular disease, which are merely results of the measuring step; and claims 5 and 16-17 recite detecting at least 3, 5, and 10 biomarkers respectively. Claims 2-7 and 16-18 fail to recite any practical applications of the recited method.

For these reasons, **claims 1-7 and 16-18** are ineligible under 35 U.S.C. 101.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-7 and 16-17 are rejected under 35 U.S.C. 102(a)(1) as being unpatentable over **Gerstein et al**. (IDS; Circulation. 2015 Dec 15; 132(24) :2297-304; a better-quality copy is uploaded) in view of **Caldwell et al**. (J Cardiovasc Pharmacol Ther. 2019 May;24(3):207-214), and as evidenced by **Dragut et al**. (Atherosclerosis, 2014. 235e239) and **Lorbeer et al**. (Heart. 2015 Feb;101(3):178-84).

Regarding **claim 1**, **Gerstein** teaches detecting, in a sample obtained from an individual, the expression level of two biomarkers NT-proBNP and angiopoietin 2 (Table 3 and pg. 2301, col. 1, par. 2).

The reference teaches analysis of the biobanked serum samples from 8494 participants with dysglycemia (pg. 2297, Background) without a healthy control group. However, the study was set to identify biomarkers of the composite outcome of myocardial infarction, stroke, or CV death. One of ordinary skill in the art would have recognized that the participants with the outcomes of myocardial infarction, stroke, or CV death were necessarily compared to the participants without the outcomes of myocardial infarction, stroke, or CV death. The participants without the outcomes of myocardial infarction, stroke, or CV death functioned as an internal control group. Therefore, **Gerstein** teaches step (c) - determining, diagnosing and/or prognosing the risk of an individual suffering from cardiovascular disease if the expression level, amount, and/or activity of the biomarker deviates from the reference from a control population.

Additionally, regarding step (c), **Gerstein** teaches that regression models were used to identify biomarkers that were each independent determinants of the composite of myocardial infarction, stroke, or CV death (pg. 2297, Methods and Results). Therefore, the identified biomarkers were "new biochemical markers of CV risk" (pg. 2297, col. 1).

Gerstein does not specifically teach decrease or increase in the expression level, amount and/or activity of NT-proBNP and angiopoietin 2. However, the reference does teach the same two biomarkers NT-proBNP and angiopoietin 2 and therefore, it inherently teaches the same changes in the expression level, amount and/or activity of the biomarkers.

Gerstein does not specifically teach administering therapeutic agents for reducing the risk of an individual from suffering from a cardiovascular disease (CVD) event.

Regarding **claim 1**, **Caldwell** teaches prevention of myocardial infarction and stroke (Title). **Caldwell** also teaches administering therapeutic agents for reducing the risk of an individual from suffering from a cardiovascular disease (CVD) event.

Specifically, **Caldwell** teaches a study with participants receiving a statin therapy, meeting the limitation of claim 1(c) reciting a statin. The participants were selected based on an elevation of high-sensitivity C-reactive protein, a sensitive marker of inflammation and accurate predictor of future risk of CVD. After 1.9 years a 44% reduction was achieved in the primary prespecified combined end point of major adverse cardiovascular events which included MI,

stroke, revascularization, hospitalization for unstable angina, and death from cardiovascular causes (pg. 210, col. 1, par. 1).

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have combined the teachings of **Gerstein** for biomarkers for cardiovascular events by employing therapeutic agents as taught by **Caldwell**, in order to reduce major adverse cardiovascular events (Caldwell, g. 210, col. 1, par. 1) as an obvious matter of combining prior art elements (diagnostics and therapy) according to known methods to yield predictable results.

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because administration of therapeutic agents does not depend on a specific method used for diagnosing a medical condition.

Regarding **claim 2**, **Gerstein** teaches glutathione S transferase alpha levels were inversely associated with all 3 studied outcomes "Although no link between an elevated glutathione S transferase a level and reduced CV outcomes has been previously reported, higher levels may be associated with reduced oxidative stress and have been noted in response to insulin administration" (pg. 2302, col. 2, par. 2). The inverse association means decrease in the transferase amount is indicative of an individual having a higher risk of suffering from cardiovascular disease or a negative prognosis, recited in claim 2.

Regarding **claim 3**, **Gerstein** does not specifically teach an increase in expression, amount, and/or activity of GHR, A2M, IGFBP2, APOB, SEPPI, TFF3, IL6 and/or CHBL1, when compared to the reference, is indicative of an individual having a higher risk of suffering from cardiovascular disease or a negative prognosis. Regarding **claim 3**, **Dragut** teaches relationship between IL-6 and cardiovascular risk in patients with type 2 diabetes and metabolic syndrome (Title). Specifically, **Dragut** provides evidence that increased plasma levels of IL-6 is associated with increased cardiovascular risk in patients with type 2 diabetes and metabolic syndrome (pg. e239, col. 1, last par.), meeting the corresponding limitation of claim 3.

Regarding **claim 4**, **Gerstein** does not specifically teach a decrease in expression, amount, and/or activity of MET, GDF15, CCL22, TNFRSF11, ANGPT2 and/or ReIA NF-KB, when compared to the reference, is indicative of an individual having a lower risk of suffering from cardiovascular disease or a positive prognosis. Regarding **claim 4**, **Lorbeer** teaches

relationship between circulating angiopoietin-2 and mortality in the general population (Title). Specifically, **Lorbeer** provides evidence that angiopoietin-2 levels were positively associated with cardiovascular mortality (pg. 1327, Methods and results and Fig. 2B), meeting the limitation of claim 4 reciting a decrease in ANGPT2 is indicative of an individual having a lower risk of suffering from cardiovascular disease or a positive prognosis.

Regarding **claim 5**, **Gerstein** teaches detecting, in a sample obtained from an individual, the expression level of at least 3 biomarkers: NT-proBNP, angiopoietin 2, and glutathione S-transferase (Table 3 and pg. 2301, col. 1, par. 2).

Regarding **claim 6**, **Gerstein** teaches the cardiovascular disease is myocardial infarction, stroke, or CV death (pg. 2297, Methods and Results) meeting the cardiovascular death, myocardial infarction, and stroke recited in claim 6.

Regarding **claim 7**, **Gerstein** teaches that the biomarkers were measured in serum samples from biobanked blood (pg. 2297, Methods and Results), meeting the limitation of claim 7 reciting blood. Serum is a derivative of blood obtained in the process of sample preparation.

Regarding **claim 16**, **Gerstein** teaches detecting, in a sample obtained from an individual, the expression level of at least 5 biomarkers: NT-proBNP, Trefoil factor 3 (TFF3), Growth differentiation factor 15 (GDF15), Apolipoprotein B (APOB), angiopoietin 2 (ANGPT2), a-2-macroglobulin (A2M), Hepatocyte growth factor receptor (MET), glutathione S-transferase (GSTA1), Tenascin C (TNC), and Selenoprotein P (SEPP1) from Table 3.

Regarding **claim 17**, **Gerstein** teaches detecting, in a sample obtained from an individual, the expression level of at least 10 biomarkers: NT-proBNP, Trefoil factor 3 (TFF3), Growth differentiation factor 15 (GDF15), Apolipoprotein B (APOB), angiopoietin 2 (ANGPT2), a-2-macroglobulin (A2M), Hepatocyte growth factor receptor (MET), glutathione S-transferase (GSTA1), Tenascin C (TNC), and Selenoprotein P (SEPP1) from Table 3.

Claim 18 is rejected under 35 U.S.C. 102(a)(1) as being unpatentable over **Gerstein** and **Caldwell**, as applied to claim 1 above, in view of **Zhao et al**. (IDS; Exp Ther Med. 2018 Feb;15(2):1456-1464).

The teachings of **Gerstein** and **Caldwell** have been set forth above.

Gerstein and **Caldwell** fail to teach the detecting comprises sequencing or performing a hybridization-based method.

Regarding **claim 18**, **Zhao** teaches "Identification of potentially relevant genes for myocardial infarction using RNA sequencing data analysis" (Title). **Zhao** also teaches the detecting comprises sequencing. Specifically, **Zhao** teaches identification of genes relevant to myocardial infarction (MI) by sequencing RNA extracted from blood samples from patients with MI and normal controls to calculate the gene expression value (Abstract). A total of eight genes, including RORA, may be considered as potentially relevant genes in the pathology of MI (pg. 1456, col. 2, par. 1). RORA is the Retinoic Acid Receptor-Related Orphan Receptor Alpha of claim 1.

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have combined the teachings of **Gerstein** for biomarkers of higher risk of suffering from cardiovascular disease with another biomarker RORA, as taught by **Zhao**, in order to add additional biomarker for diagnosing and/or prognosing an individual's risk of suffering from cardiovascular disease.

MPEP 2144.06 provides "[i]t is prima facie obvious to combine two compositions each of which is taught by the prior art to be useful for the same purpose, in order to form a third composition to be used for the very same purpose.... [T]he idea of combining them flows logically from their having been individually taught in the prior art." In re Kerkhoven, 626 F.2d 846, 850, 205 USPQ 1069, 1072 (CCPA 1980).

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because each biomarker is measured using its own specific set of reagents and detection of one biomarker does not affect detection of another.

Response to Arguments

Applicant's arguments filed August 13, 2026 have been fully considered.

Claims 1-7 and 16-18 were rejected under 35 U.S.C. § 112(a) as failing to comply with the written descript requirement.

Applicant argues that "specification fully describes in detail the methods used to identify the biomarkers recited in the claims and to correlate a decrease, or increase, in the expression, amount, and/or activity of the biomarker with a higher risk of suffering from a cardiovascular disease (CVD) event" (pg. 7, last par.). Applicant further argues that the biomarkers were disclosed in the specification and that "As described at page 101, paragraph 7, 13 protein

biomarkers (a subset of those recited in amended claim 1) and genotypes from 2 GWAs genes were used to run the paradigm “, “13 protein biomarkers are listed at page 102 “, “FIG. 5 is described in detail at page 92”, and “Based on the detailed description on the workflow and results provided in Example 1 and FIG. 5, a person of ordinary skill in the art would understand that the inventors identified the recited biomarkers and evidence of correlations with cardiovascular outcomes (pg. 8, par. 1).

The arguments are not persuasive because Applicant still fails to identify sections of the specification disclosing specific data as stated in the rejection (non-final OA, mailed May 14, 2026; pg. 4, par. 3): “*The specification is silent with respect to specific data related to recited biomarkers and evidence of correlations with cardiovascular outcomes. For example, the specification fails to provide any evidence of corresponding p-values, which is the key parameter characterizing a biomarker for its suitability in diagnostic applications. No biomarkers and corresponding p-values are disclosed. No combinations of the biomarkers and corresponding p-values for these combinations are disclosed.*” The disclosure of the biomarkers by their names is insufficient to demonstrate possession of the invention claiming measuring expression level, amount, and/or activity of the biomarkers - the specification fails to disclose results of these measurements.

Example 1 discloses identification of biomarkers and biomarker networks but fails to disclose measurements of the expression level, amount, and/or activity of the biomarkers.

Table presented in Fig. 5 discloses hazard ratios (HR; right column) but still fails to disclose measurements expression level, amount, and/or activity of the biomarkers. A hazard ratio is recognized in the art as a measure from survival analysis that compares the instantaneous event rate between two groups; an HR of 1 means equal hazard, values above 1 indicate higher hazard, and values below 1 indicate lower hazard in the treatment or exposed group. As such, a hazard ratio is not a direct substitute for measurements of the expression level, amount and/or activity of the recited biomarkers.

Applicant further argues that “Applicant submits that various methods of measuring the expression, amount, and/or activity of the recited biomarkers was well known in the art and that a person of ordinary skill in the art would conclude that the inventors has possession of a method including measuring the recited biomarkers at the time of filing. Further, Applicant's

specification fully describes several methods of detecting the expression level, amount, and/or activity of the recited biomarkers. See, for example, Applicant's specification at pages 4-7" (pg. 8, par. 2).

The argument is not persuasive because a description of "several methods of detecting the expression level, amount, and/or activity" on pages 4-7 of the specification is not a substitute for disclosing actual data linking measured expression level, amount, and/or activity of the biomarkers with CVD events.

Claims 1-7 and 16-18 were rejected under 35 U.S.C. § 101 as being directed to a naturally occurring correlation without significantly more.

Applicant argues that amended claim 1 recites an additional step of "administering to the individual a therapeutic agent selected from a statin, beta blocker, blood thinning agent, nitrates, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists, calcium channel blockers and diuretics, thereby reducing the risk of the individual from suffering from a CVD event" (pg. 9 last par. – pg. 10, par. 1-2).

The argument is not persuasive because claim 1 the recited therapeutic agents: statin, beta blocker, blood thinning agent, nitrates, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists, calcium channel blockers and diuretics are not intended as universal treatments to treat any cardiovascular disease events. Therefore, the recited treatments are not particular. For example, administration of blood thinning agents to a patient with stroke caused by a burst of a blood vessel is going to harm a patient. The specification fails to disclose biomarkers capable of differentiating different cardiovascular disease events. Bundling all cardiovascular disease events into one CVD group fails to make administration of the recited therapeutic agents a practical application of the disclosed diagnostic method due to mutually exclusive medications required to treat different cardiovascular disease events.

Additionally, Applicant argues that the administration step makes the combination of steps not routine and conventional (pg. 10, par. 2). The argument is not persuasive because a combination of a diagnostic step and a drug administering step is routine and conventional as evidenced by current healthcare routines. Applicant fails to provide evidence why instant combination is not routine and conventional.

Claims 1-7 and 16-18 were rejected under 35 U.S.C. § 103 as unpatentable over the cited prior art references.

Applicant argues that the prior art does not teach or suggest administering to an individual a therapeutic agent selected from a statin, beta blocker, blood thinning agent, nitrates, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists, calcium channel blockers and diuretics, as required by amended claim 1 (pg. 11, par. 2). The argument is persuasive. The rejections of claims 1-7 and 16-18 under 35 U.S.C. 103 are withdrawn in view of claim 1 amendments. Claim 1 amendments necessitated a new prior art search and new grounds of rejection have been found in view of newly found prior art reference. The amended claim 1 is rejected under 35 U.S.C. § 103 as unpatentable over **Gerstein** in view of **Caldwell**.

Briefly, **Caldwell** teaches prevention of myocardial infarction and stroke (Title). Specifically, **Caldwell** teaches a study with participants receiving a statin therapy and achieving a 44% reduction in the primary prespecified combined end point of major adverse cardiovascular events which included MI, stroke, revascularization, hospitalization for unstable angina, and death from cardiovascular causes (pg. 210, col. 1, par. 1).

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Alexander Volkov whose telephone number is (571) 272-1899. The examiner can normally be reached M-F 9:00AM-5:00PM (EST).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached on (571) 272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from Patent Center. Status information for published applications may be obtained from Patent Center. Status information for unpublished applications is available through Patent Center for authorized users only. Should you have questions about access to Patent Center, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) Form at https://www.uspto.gov/patents/uspto-automated- interview-request-air-form.

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